AP Electric Industrial · AP Rack RL4 rev A (CONCEPTO): TCA9554 (0x20) –> 74HCT125 –> 4 x NCV8406A –> 4 x relé G2RL-1A-E DC24,
8 entradas de 12–30 V LTV-217 (TCA9554 0x22, 0x23 LED), SP3485 + 2 x JST XH4 (AP LINK), ESP32-C3. Alimentación 24 V. Sin red de CA.

PWR_FLAG VIN_RAW
PWR_FLAG V24
PWR_FLAG V5
PWR_FLAG 3V3
PWR_FLAG COM
PWR_FLAG GND

VIN_RAW 1 — 2 GND
J1
VIN 12–30V

VIN_RAW
F1
MINI 3A
V24

V24
D1
SMBJ33A
GND

V24
C1
10uF 50V
GND

V24
C2
10uF 50V
GND

V24 5 VIN — CB 1 BST
V24 4 SHDN — SW 6 SW
GND — FB 3 FB
2 GND
U1
LMR16006XDDC

V5
L1
47uH
SW

SW
D2
SS210
GND

BST
C3
100nF
SW

V24
C4
10uF 50V
GND

V5
C5
22uF 25V
GND

V5
RB1
56k
FB

FB
RB2
10k
GND

V5
D3
B5819W
VBUS

V5 3 IN — SW 5 SWP
V5 2 EN — BST 6 BSTP
GND — FB 1 3V3
4 GND
U2
AP63203WU

SWP
L2
4.7uH
3V3

BSTP
C6
100nF
SWP

V5
C7
10uF 50V
GND

3V3
C8
22uF
GND

3V3
C9
22uF
GND

3V3
1 3V3
EN 2 EN — IO20/RXD 11
IO21/TXD 12
VSENSE 18 IO0
RS_TX 17 IO1
SDA 16 IO2
RS_RX 15 IO3
INT 3 IO4
SENSE_I 4 IO5
SENSE_O 5 IO6
6 IO7
SCL 7 IO8
BOOT 8 IO9
RS_DE 10 IO10
USB_DN 13 IO18
USB_DP 14 IO19
GND — GND
9 GND
U3
ESP32-C3-WROOM-02

3V3
C10
10uF
GND

3V3
C11
100nF
GND

3V3
R1
10k
EN

EN
C12
1uF
GND

3V3
R2
10k
BOOT

BOOT SW1 GND
MODO

EN SW2 GND
RESET

3V3
R3
4.7k
SDA

3V3
R4
4.7k
SCL

3V3
R5
10k
INT

V24
R6
100k
VSENSE

VSENSE
R7
10k
GND

VSENSE
C13
100nF
GND

VBUS A4 VBUS
CC1 A5 CC1U
CC2 B5 CC2U
D– A7 USB_DN
D– B7 USB_DN J2
D+ A6 USB_DP USB-C
D+ B6 USB_DP
SHIELD — GND — SBU1 A8
SBU2 B8
S1 — A1
GND — 3V3GND
C14
100nF
GND

CC1U
R8
5.1k
GND

CC2U
R9
5.1k
GND

VBUS
5
USB_DN 1 — 6 USB_DN
USB_DP 3 — 4 USB_DP
2 GND
U4
USBLC6-2SC6

SENSE_IC 1 RSB 2 RSA 3 GND 4
JL1
AP LINK ENTRA

SENSE_OC 1 RSB 2 RSA 3 GND 4
JL2
AP LINK SIGUE

3V3
8 VCC
RS_RX 1 RO — A 6 RSA
RS_DE 2 RE — B 7 RSB
RS_DE 3 DE
RS_TX 4 DI
GND — GND
5 GND
U5
SP3485EN

RS_DE
R10
10k
GND

RSA 1 — 2 RSB
3
GND
D4
SM712

RSA
R11
120
TERM

TERM JP1 RSB
120R

PA
R12
680
RSA

RSB
R13
680
PB

3V3 JP2 PA
POL A

Estado: concepto. Límites del producto por determinar por ensayo
Placa de 2 capas, 2 oz (JLCPCB)
PCB 176 x 100 mm, 2 capas, 2 oz

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Size: A3	Date: 2026-10-16	**Rev: 1.0**
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